

Cartwheel Homework 3

Scenario design, pilot review, and final trace export

250 final scenarios	250 completed runs	260 exported traces	5 pilot failures
---	--	---	--

Outcome

We designed a grounded scenario space, ran and reviewed a 30-scenario pilot, confirmed five useful agent failures, expanded the dataset to 250 reviewed scenarios, completed every run on one model, and exported a trace bundle covering every final scenario identifier.

Commit: **5b0266b** - Model: **gpt-5.6-luna** - Service tier: **Fast** - Report date: **September 17, 2026**

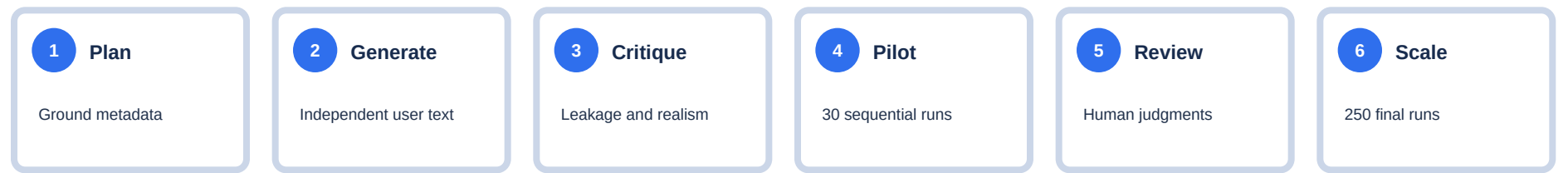
Screenshots were captured from the local Langfuse project after the final run and show the actual traces included in the committed export.

1. Evaluation design

The scenario plan varied seven dimensions while keeping expected behavior grounded in SQLite rows, deterministic eligibility rules, policy documents, specification requirements, and the documented damaged-record table.



Workflow



2. Pilot composition and review

Balanced pools

15 coverage and 15 challenge scenarios. All three roles and all eight user styles appeared in both the design and validation audit.

Damaged records

All six documented data-quality cases appeared exactly once, including reversed dates, missing dates, store mismatch, duplicate title, invalid price, and missing title.

Human review

12 Langfuse results were reviewed. Every scenario was valid; five observed behaviors were confirmed as agent failures.

Pilot check	Result
Runner completion	30 / 30 completed; zero execution errors
Review set	12 distinct scenarios
Confirmed failures	5 valid-scenario failures
Model consistency	gpt-5.6-luna, Fast tier
Leakage audit	No tool, trace, prompt, metadata, or requirement-label leakage

The pilot did its job: ordinary requests mostly worked, while difficult cases exposed specific, evidence-backed behavior to study in Homework 4.

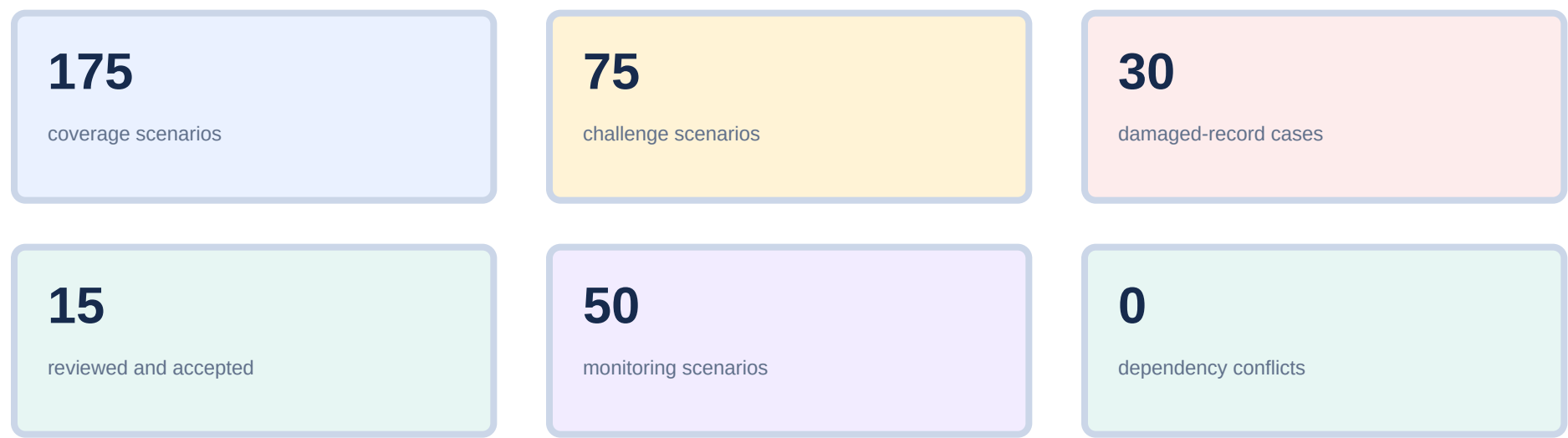
3. What the pilot revealed

Scenario	Observed behavior	Evidence supporting the judgment
pilot-0002	Reported 20 orders in total .	The tool returned its capped 20-row page; user 1 had 26 seeded orders.
pilot-0003	Selected order 4455 despite unresolved ambiguity.	The fuzzy search returned three matching delivered vase orders.
pilot-0008	Returned a 20-row, nine-column merchant table and aggregate counts.	The expected criterion required a concise answer; store 2 had 542 seeded orders, so page counts could mislead.
pilot-0027	Answered Blue Heron Ceramics without identifying inconsistent data.	Order 8003 is stored under store 1 while product 553 belongs to store 14.
pilot-0029	Affirmed availability and exposed a -\$5.00 catalog price.	Product 4 contains an invalid negative price and should not be presented as a valid offer.

Interpretation

The confirmed failures were grounded in tool limits, database values, damaged records, or explicit response requirements. We did not count stylistic preferences alone, and we did not turn the pilot findings into a taxonomy during Homework 3.

4. Final scenario set



Review coverage: the 15-scenario human sample included both scenario groups, all three roles, and one representative of every intent. All 15 were accepted with no changes.

Monitoring subset: 50 final records were selected for later comparison: 35 coverage, 15 challenge, all three roles, and all 15 intents.

State safety: every eligible cancellation and refund used a unique order. A dependency audit found no later explicit order scenario whose expected state was invalidated by an earlier write.

Strict validator: 250 unique identifiers, exact group counts, turn-count consistency, no duplicate conversations, and exactly five final scenarios for each documented damaged record.

5. Final run and trace export

175/175

coverage completed

75/75

challenge completed

0

runner errors

250/250

scenario IDs exported

Tool	Calls
get_order	173
search_products	88
search_help_center	74
get_policy	60
escalate_to_human	31
list_my_orders	28
find_order	20
issue_refund	15
cancel_order	15
get_my_store	2

Trace bundle

260 traces
250 scenario identifiers
10 extra traces for second turns
674 model generations

Observed model usage

604,627 input tokens
85,386 output tokens
\$1.13551 calculated cost

Verification

Three exported traces were inspected directly: a routine coverage lookup, a damaged-record challenge, and a two-turn dispute. Each contained conversation input/output, model name, tool activity, and the expected `cartwheel_scenario_id`.

6. Coverage trace

Final scenario **support-0001** shows the full conversation, the **get_order** branch, model cost and latency, authenticated shopper metadata, prompt version, and the stable scenario identifier.

langfuse v3.225.8 OSS

Cartwheel Course / Cartwheel Dev / Traces

Go to...

Home

Dashboards

Observability

Tracing

Sessions

Users

Prompt Management

Prompts

Playground

Evaluation

Scores

Evaluators

Human Annotation

Datasets

Star Langfuse

See the latest releases and help grow the community on GitHub

Langfuse 35k

Settings

Book a call

Support

Student student@exampl...

Trace cartwheel.session_message: b93088955b6d0bd9b63f004d713c4a4e

Search

Aggregated Expanded

Graph

cartwheel.session_message 5.27s \$0.002839

cartwheel.session_message 5.27s \$0.002839

Agent Workflow 5.26s \$0.002839

cartwheel-support.agent 5.25s \$0.002839

openai.response 2.45s 786 → 64 (Σ 850) \$0.00117

get_order 0.01s

openai.response 2.78s 967 → 117 (Σ 1,084) \$0.001669

cartwheel.session_message :

2026-09-17 10:06:19.652

Latency: 5.27s Env: default \$0.002839 1,753 prompt → 181 completion (Σ 1,934)

Preview Log View Scores

User

Hi, could you please check the current status of my order 2?

Assistant

Order 2 from Trailhead Supply was delivered on December 24, 2025.

Ordered: December 19, 2025

Shipped: December 21, 2025

Total: \$217.75

Corrected Output

Click to add corrected output

Metadata

Path	Value
attributes	5 items
cartwheel.session_id	"eb7f8cae-1174-4ab3-99c9-376106b8eb62"
cartwheel.user_role	"shopper"
cartwheel.user_id	"129"
cartwheel.prompt_version	"f4cf9d1978a4"
cartwheel.scenario_id	"support-0001"
resourceAttributes	5 items
telemetry.sdk.language	"python"
telemetry.sdk.name	"opentelemetry"
telemetry.sdk.version	"1.43.0"
service.instance.id	"580061ec-57f1-433c-8100-fca93b8a05f9"
service.name	"unknown_service"
scope	2 items
name	"cartwheel.server"
attributes	empty object

langfuse

v3.225.8 OSS

Home

Dashboards

Observability

Tracing

Sessions

Users

Prompt Management

Prompts

Playground

Evaluation

Scores

Evaluators

Human Annotation

Datasets

Settings

Book a call

Support

Student
student@exampl...

Cartwheel Course

Cartwheel Dev

Traces

Trace cartwheel.session_message: 12c3e40886be28944cfc6625a5d00526

Search

Aggregated

Expanded

cartwheel.session_message

7.56s ⚡ \$0.00593

cartwheel.session_message

7.56s ⚡ \$0.00593

Agent Workflow

7.55s ⚡ \$0.00593

cartwheel-support.agent

7.55s ⚡ \$0.00593

openai.response

1.46s 786 → 64 (Σ 850) \$0.00117

get_order

0.00s

openai.response

3.04s 969 → 304 (Σ 1,273) \$0.002793

escalate_to_human

0.00s

openai.response

3.02s 1,307 → 110 (Σ 1,417) \$0.001967

escalate_to_human

2026-09-17 10:21:57.166

Latency: 0.00s Env: default

Preview Scores

Formatted JSON

Input

Path	Value
summary	"Verify inconsistent shipping and delivery dates for order 8001"
context	"Shopper 174 asked for the shipping and delivery timeline for order 8001 from Harbor Knits. The order record shows status delivered, delivered_at 2026-06-23, and shipped_at 2026-06-25; the shipment date is later than the delivery date. Please verify the correct timeline and follow up with the shopper."

Output

Path	Value
ok	true
ticket_id	157
sla_hours	24

Corrected Output

Click to add corrected output

Metadata

Path	Value
attributes	8 items
traceloop.span.kind	"tool"
gen_ai.tool.name	"escalate_to_human"
gen_ai.tool.type	"function"
gen_ai.provider.name	"openai"
gen_ai.operation.name	"execute_tool"
cartwheel.user_role	"shopper"
cartwheel.user_id	"174"
cartwheel.permission_denied	false
resourceAttributes	5 items
telemetry.sdk.language	"python"
telemetry.sdk.name	"opentelemetry"
telemetry.sdk.version	"1.43.0"
service.instance.id	"580061ec-57f1-433c-8100-fca93b8a05f9"

Graph

Aggregated

Expanded

__start__

cartwheel.session_message

Agent Workflow

cartwheel-support.agent

openai.response (3/3)

__end__

get_order

escalate_to_human

8. Multi-turn dispute: initial request

The first trace for **support-0241** contains the disputed order lookup, escalation path, ticket 177, response, and shared session identifier used to connect the conversation turns.

langfuse v3.225.8 OSS

Cartwheel Course / Cartwheel Dev / Traces

Go to...

Home

Dashboards

Observability

Tracing

Sessions

Users

Prompt Management

Prompts

Playground

Evaluation

Scores

Evaluators

Human Annotation

Datasets

Star Langfuse

See the latest releases and help grow the community on GitHub

Langfuse 35k

Settings

Book a call

Support

Student student@exampl...

Trace cartwheel.session_message: 5c72cdd8bf246932d65702d0b8b9b18d

Search

Aggregated Expanded

Graph

cartwheel.session_message 6.50s \$0.005575

cartwheel.session_message 6.50s \$0.005575

Agent Workflow 6.50s \$0.005575

cartwheel-support.agent 6.49s \$0.005575

openai.response 2.13s 788 → 95 (\$ 883) \$0.001358

get_order 0.01s

openai.response 2.83s 1,000 → 233 (\$ 1,233) \$0.002398

escalate_to_human 0.01s

openai.response 1.50s 1,267 → 92 (\$ 1,359) \$0.001819

cartwheel.session_message : 2026-09-17 10:30:50.392

Latency: 6.50s Env: default \$0.005575 3,055 prompt → 420 completion (\$ 3,475)

Preview Log View Scores

User

The charge on order 145 is incorrect, and I'd like to dispute it.

Assistant

I found order 145 from Paper Lantern Press for \$156.75. It was delivered on March 25, 2025, and is marked not eligible for a standard refund.

Because I can't directly process a payment-card dispute, I escalated this to a human support agent for review. Your ticket is #177. A human will follow up within approximately 24 hours.

Corrected Output

Click to add corrected output

Metadata

Path	Value
attributes	5 items
cartwheel.session_id	"8ea9faf9-daa0-42f2-8dc1-2dac83f2bf2c"
cartwheel.user_role	"shopper"
cartwheel.user_id	"1"
cartwheel.prompt_version	"f4cf9d1978a4"
cartwheel.scenario_id	"support-0241"
resourceAttributes	5 items
telemetry.sdk.language	"python"
telemetry.sdk.name	"opentelemetry"
telemetry.sdk.version	"1.43.0"
service.instance.id	"580061ec-57f1-433c-8100-fca93b8a05f9"
service.name	"unknown_service"
scope	2 items
name	"cartwheel.server"
attributes	empty object

9. Multi-turn dispute: follow-up

The second trace preserves the same `cartwheel.session_id` and `cartwheel.scenario_id`. The reply correctly refers to the existing ticket rather than creating another escalation.

langfuse v3.225.8 OSS

Cartwheel Course / Cartwheel Dev / Traces

Go to...

Home

Dashboards

Observability

Tracing

Sessions

Users

Prompt Management

Prompts

Playground

Evaluation

Scores

Evaluators

Human Annotation

Datasets

Star Langfuse

See the latest releases and help grow the community on GitHub

Langfuse 35k

Settings

Book a call

Support

Student student@exampl...

Trace cartwheel.session_message: e4fe11a7cb81181cc2ef96ff31ca9122

Search

cartwheel.session_message 3.21s \$0.000851

cartwheel.session_message 3.21s \$0.000851

Agent Workflow 3.21s \$0.000851

cartwheel-support.agent 3.21s \$0.000851

openai.response 3.19s 1,377 → 102 (\$1,479) \$0.000851

Graph

Aggregated Expanded

Start

cartwheel.session_message

Agent Workflow

cartwheel-support.agent

openai.response

End

cartwheel.session_message :

2026-09-17 10:30:56.893

Latency: 3.21s Env: default \$0.000851 1,377 prompt → 102 completion (\$1,479)

Preview Log View Scores

User

I'd like this handled by a human representative, please.

Assistant

Certainly. Your request has already been escalated to a human representative under ticket #177. They will review the disputed \$156.75 charge on order 145 and follow up within approximately 24 hours.

Corrected Output

Click to add corrected output

Metadata

Path	Value
attributes	5 items
cartwheel.session_id	"8ea9faf9-daa0-42f2-8dc1-2dac83f2bf2c"
cartwheel.user_role	"shopper"
cartwheel.user_id	"1"
cartwheel.prompt_version	"f4cf9d1978a4"
cartwheel.scenario_id	"support-0241"
resourceAttributes	5 items
telemetry.sdk.language	"python"
telemetry.sdk.name	"opentelemetry"
telemetry.sdk.version	"1.43.0"
service.instance.id	"580061ec-57f1-433c-8100-fca93b8a05f9"
service.name	"unknown_service"
scope	2 items
name	"cartwheel.server"
attributes	empty object

10. Findings and recommendations

Ground truth must stay separate

Scenario prompts remained natural while expected behavior cited database rows, policies, code rules, or the specification.

Challenge data changes observed quality

Coverage and challenge counts must remain separate because the challenge pool intentionally concentrates boundary and damaged-data cases.

Pagination creates semantic risk

A capped list can be accurate as a page yet wrong when described as the user's complete history.

Data corruption needs explicit handling

The strongest failures occurred when the agent repeated malformed values or selected one side of an inconsistent record.

Session grouping matters

A multi-turn conversation produced one trace per user turn; Homework 4's interface must group traces by session before review.

Recommendation for Homework 4: begin open coding with a mixed sample that includes ordinary coverage, access denials, write actions, damaged records, and multi-turn sessions. Review the earliest consequential failure in context before proposing named failure modes.